

PYTHON PROJECT CODE

```
import pandas as pd
import matplotlib.pyplot as plt

# -----
# DATA HANDLING MODULE
# -----

def load_data(file_path):
    try:
        data = pd.read_csv(file_path)
        print("Data loaded successfully")
        return data
    except Exception as e:
        print("Error loading data:", e)
        return None

def clean_data(df):
    print("Cleaning data...")
    return df.dropna()

# -----
# AQI CALCULATION MODULE
# -----

def calculate_aqi(row):
```

```
pm25 = row['PM2.5']
```

```
pm10 = row['PM10']
```

```
co = row['CO']
```

```
# Simple AQI formula (demo)
```

```
aqi = (pm25 * 0.5) + (pm10 * 0.3) + (co * 0.2)
```

```
return round(aqi, 2)
```

```
def add_aqi(df):
```

```
    df['AQI'] = df.apply(calculate_aqi, axis=1)
```

```
    return df
```

```
# -----
```

```
# ANALYSIS MODULE
```

```
# -----
```

```
def basic_stats(df):
```

```
    print("\n--- AQI Statistics ---")
```

```
    print("Average AQI:", round(df['AQI'].mean(), 2))
```

```
    print("Max AQI:", df['AQI'].max())
```

```
    print("Min AQI:", df['AQI'].min())
```

```
def pollution_level(aqi):
```

```
    if aqi <= 50:
```

```
        return "Good"
```

```
    elif aqi <= 100:
```

```
        return "Moderate"
```

```
elif aqi <= 150:
    return "Unhealthy for Sensitive Groups"
elif aqi <= 200:
    return "Unhealthy"
else:
    return "Very Unhealthy"
```

```
# -----
```

```
# VISUALIZATION MODULE
```

```
# -----
```

```
def plot_aqi(df):
    plt.plot(df['AQI'])
    plt.title("AQI Trend")
    plt.xlabel("Index")
    plt.ylabel("AQI")
    plt.show()
```

```
def bar_pollutants(df):
    avg = df[['PM2.5','PM10','CO']].mean()
    avg.plot(kind='bar')
    plt.title("Average Pollutant Levels")
    plt.show()
```

```
# -----
```

```
# MAIN PROGRAM
```

```
# -----
```

```
def main():  
    file_path = "air_quality_data.csv"  
  
    data = load_data(file_path)  
    if data is None:  
        return  
  
    data = clean_data(data)  
    data = add_aqi(data)  
  
    basic_stats(data)  
  
    print("\n--- Pollution Levels ---")  
    for i in range(len(data)):  
        aqi = data.loc[i, 'AQI']  
        print(f"Row {i}: AQI={aqi} → {pollution_level(aqi)}")  
  
    # Graphs  
    plot_aqi(data)  
    bar_pollutants(data)  
  
    # Run program  
if __name__ == "__main__":  
    main()
```